Private Teacher Script — SI Units and Unit Conversion

Duration: 30 minutes

Audience: New Class 11 students

Important: This is a private presenter script. Keep it on the laptop/second screen. Show students only `SI-Units-and-Unit-Conversion-clean.pptx`.

Before the lecture

- Open the slides in PowerPoint and use Presenter View.
- Keep a pen/stylus ready for derivations and student working.
- Do not rush to give answers. Wait 5–8 seconds after each question.
- Prefer the NCERT spelling metre while speaking.
- Main learning outcome: Students should understand that a physical quantity needs a numerical value and a unit, recall the seven SI base units, and perform basic speed conversions.

Slide 1 — SI Units and Unit Conversion

Time: 0:00–1:30

Say

“Good morning. Before we begin, let me ask you something. If I tell you that the length of this table is two, have I given you complete information?”

Pause and allow students to respond.

“Two what—metres, centimetres, feet? The number is present, but the measurement is incomplete. Today we will understand why units are essential and learn how to convert one unit into another without memorising blindly.”

“By the end of this short class, you should be able to explain what makes a measurement meaningful, identify SI base units, and convert speed between kilometres per hour and metres per second.”

Teacher action

Write only:

```
Physical quantity = numerical value × unit
```

Transition

“Let us test this idea with a familiar example.”

Slide 2 — Hook Question

Time: 1:30–4:00

Say

“Suppose someone says, ‘The speed of a car is 20.’ Is that statement complete?”

Wait for answers.

“What could 20 mean? It might be 20 kilometres per hour, 20 metres per second, or even 20 miles per hour. These values describe very different speeds.”

“Therefore, a number alone does not fully describe a physical quantity. We need both a numerical value and a standard unit.”

Ask

1. “Which is faster: 20 m/s or 20 km/h?”
2. “Can we compare them directly?”

Expected response

Students may guess that 20 m/s is faster. Confirm that comparison requires conversion to the same unit.

Say

“This is exactly why unit conversion matters: quantities can be compared or used in an equation only when their units are compatible.”

Transition

“First, let us define what a measurement actually contains.”

Slide 3 — What Is a Measurement?

Time: 4:00–7:00

Say

“Measurement means comparing an unknown physical quantity with an accepted standard of the same kind.”

“Every measured physical quantity has two parts:

1. a numerical value, which tells us how much; and
2. a unit, which tells us the standard being used.”

“For example, in 5 metres, 5 is the numerical value and metre is the unit.”

Ask

“In the measurement 10 seconds, what is the numerical value and what is the unit?”

Expected response

Numerical value = 10; unit = second.

Clarify

“A physical quantity is something that can be measured and expressed using a number and a unit—for example length, mass, time, speed and force.”

Board work

Write:

`Q = n × u`

Explain:

- `Q` = physical quantity
- `n` = numerical value
- `u` = unit

Transition

“Scientists need common standards so that a measurement means the same thing everywhere. That common system is SI.”

Slide 4 — The Seven SI Base Units

Time: 7:00–12:00

Say

“SI stands for the International System of Units. It provides one internationally accepted unit system.”

“There are seven base quantities. Their units are called base units because they are independent and are not defined by combining other physical units.”

Read the table steadily:

- Length — metre — `m`
- Mass — kilogram — `kg`
- Time — second — `s`
- Thermodynamic temperature — kelvin — `K`
- Electric current — ampere — `A`
- Luminous intensity — candela — `cd`
- Amount of substance — mole — `mol`

Important points to say

“Unit symbols are case-sensitive. Capital and small letters are not interchangeable.”

“For example, `K` is the symbol for kelvin and `A` is the symbol for ampere.”

“Symbols are not pluralised. We write 5 kg, not 5 kgs, and we do not place a full stop after a symbol unless it ends the sentence.”

Quick recall

Ask students:

1. “What is the SI unit of mass?” — kilogram
2. “What is the symbol of ampere?” — A
3. “Which base quantity uses mole?” — amount of substance
4. “Is gram the SI base unit of mass?” — No, kilogram is the base unit

Transition

“Base units are our building blocks. Other units are produced by combining them.”

Slide 5 — Derived Units

Time: 12:00–15:00

Say

“A derived physical quantity is defined using one or more base quantities. Its unit is therefore formed by combining base units.”

“Speed is distance divided by time, so its SI unit is metre per second, m/s.”

“Force equals mass multiplied by acceleration:

$F = ma$

Mass has unit kilogram and acceleration has unit metre per second squared. Therefore, the SI unit of force is $kg \cdot m/s^2$. We give this combination the special name newton, symbol N.”

Briefly explain:

- Energy — joule (J) — $kg \cdot m^2/s^2$
- Power — watt (W) — $kg \cdot m^2/s^3$

Ask

“Is metre per second a base unit or a derived unit?”

Expected response

Derived unit, because it combines metre and second.

Transition

“Now let us see why the same quantity may appear in different units.”

Slide 6 — Why Convert Units?

Time: 15:00–17:00

Say

“In daily life, vehicle speed is usually written in kilometres per hour. In physics equations, speed is often required in metres per second.”

“Before substituting values into an equation, all quantities should be converted into a consistent unit system—normally SI units.”

“A correct formula can still produce a wrong answer if incompatible units are substituted.”

Example to speak

“If distance is in metres but time is in hours, writing the result as metres per second would be incorrect unless hours are first converted to seconds.”

Transition

“Let us derive the conversion instead of memorising it.”

Slide 7 — Converting km/h to m/s

Time: 17:00–21:00

Say and write step by step

“Start with one kilometre per hour.”

Write:

`1 km/h`

“We know that 1 kilometre equals 1000 metres and 1 hour equals 3600 seconds.”

Write:

`1 km/h = 1000 m / 3600 s`

“Now simplify 1000 divided by 3600.”

Write:

`1 km/h = 10/36 m/s = 5/18 m/s`

“Therefore, to convert kilometres per hour into metres per second, multiply by `5/18`.”

Check understanding

Ask:

“If a speed is given in km/h, should its numerical value become larger or smaller when converted to m/s?”

Expected response: smaller.

“Correct, because one metre per second is a faster unit than one kilometre per hour, so fewer m/s units represent the same speed.”

Transition

“The reverse conversion follows by taking the reciprocal.”

Slide 8 — Converting m/s to km/h

Time: 21:00–24:00

Say and write

“From the previous result:

`1 km/h = 5/18 m/s`

Taking the reciprocal relation gives:

`1 m/s = 18/5 km/h = 3.6 km/h`.”

“Therefore:

- km/h to m/s: multiply by `5/18`
- m/s to km/h: multiply by `18/5`, or 3.6”

Memory aid

“When moving from km/h to m/s, the number usually becomes smaller. When moving from m/s to km/h, it becomes larger.”

Oral check before the next slide

Ask without showing the answer:

“What is 5 m/s in km/h?”

Expected working:

`5 × 18/5 = 18 km/h`

Transition

“Now you will solve two conversions.”

Slide 9 — Practice

Time: 24:00–27:00

Important: The slide contains the solutions. Ask both questions orally before moving to this slide. Then use the slide only to confirm the answers.

Question 1

“Convert 72 km/h into m/s.”

Give students 20–30 seconds.

Expected working:

```
72 × 5/18 = 20 m/s
```

Question 2

“Convert 10 m/s into km/h.”

Expected working:

```
10 × 18/5 = 36 km/h
```

Ask

“How can we quickly check whether these answers are reasonable?”

Expected response

- km/h to m/s should produce a smaller numerical value.
- m/s to km/h should produce a larger numerical value.

Address a common mistake

“Do not use 5/18 in both directions. Identify the starting unit and the required unit before choosing the factor.”

Transition

“Units can also help us test whether a physics equation is possible.”

Slide 10 — Dimensional Analysis

Time: 27:00–29:00

Say

“This is only a short preview; dimensional analysis will be studied in detail later.”

“A physically meaningful equation must have the same dimensions on both sides.”

“For force:

$F = ma$

The unit on the right is:

$kg \times m/s^2 = kg \cdot m/s^2$

This is the unit of force, newton. So the equation is dimensionally consistent.”

Important caution

“Dimensional consistency is a necessary check, but it does not prove that an equation is completely correct. A dimensionally correct equation may still have an incorrect numerical constant.”

Transition

“Let us finish by returning to our opening question.”

Slide 11 — Exit Question

Time: 29:00–30:00

Ask

“Why can we not simply say that the speed of a car is 20?”

Wait for one complete sentence.

Model answer

“Because a physical quantity is meaningful only when its numerical value is accompanied by an appropriate unit; therefore, the speed must be stated as something like 20 m/s or 20 km/h.”

Close

“Today we learned that units are not labels added at the end—they are part of the physical quantity itself. We also derived the conversion factors rather than memorising them. In the next class, we will use units and dimensions to test equations and solve more numerical problems.”

Emergency shorter version

If only 20 minutes are available:

- Slides 1–3: 4 minutes
- Slide 4: 4 minutes
- Slide 5: 2 minutes
- Slides 6–8: 6 minutes
- Slide 9: 3 minutes
- Skip Slide 10
- Slide 11: 1 minute

Common questions students may ask

“Why is kilogram a base unit instead of gram?”

The SI base unit of mass was historically defined as the kilogram. Prefixes are normally attached to units, but kilogram is the special base-unit exception.

“Why do we multiply by 5/18?”

Because `1 km = 1000 m` and `1 h = 3600 s`, so:

`1000/3600 = 5/18`.

“Can units be cancelled like numbers?”

Yes. Conversion factors are ratios equal to one, so unwanted units cancel algebraically while the physical quantity remains unchanged.

“Does dimensional correctness guarantee that a formula is correct?”

No. It detects many errors, but it cannot determine dimensionless numerical constants such as 2, 1/2 or π.